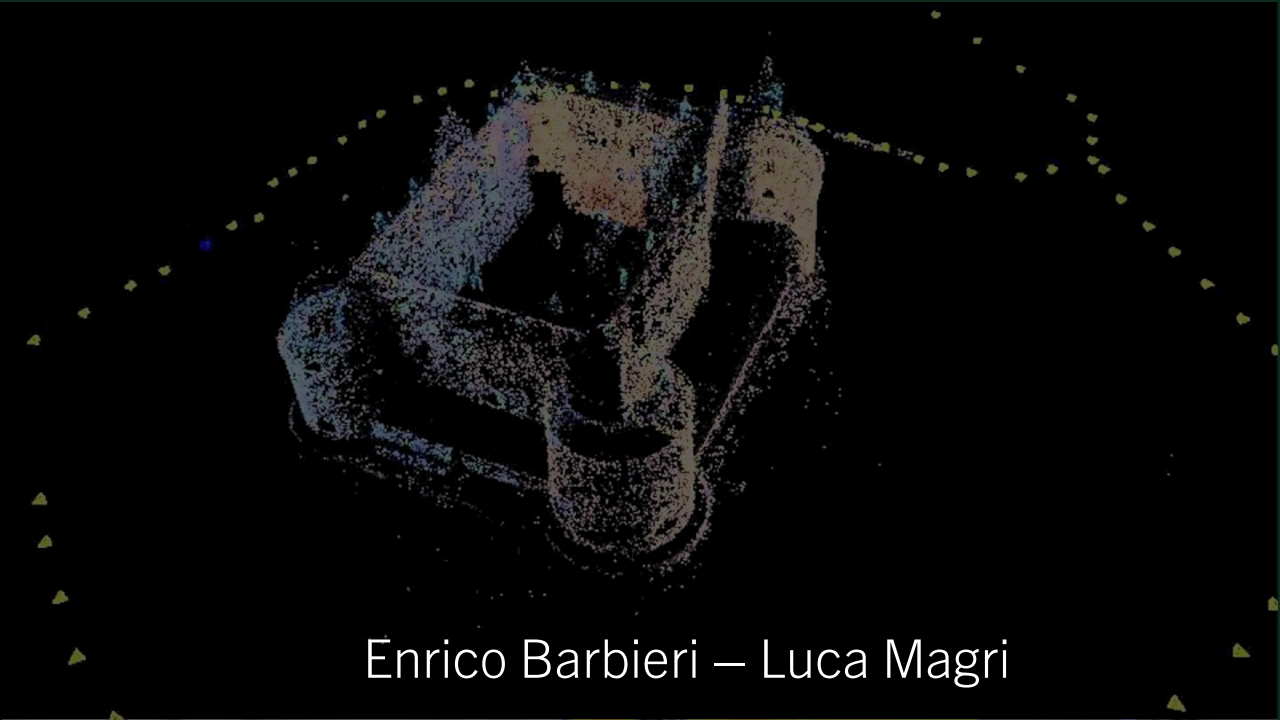
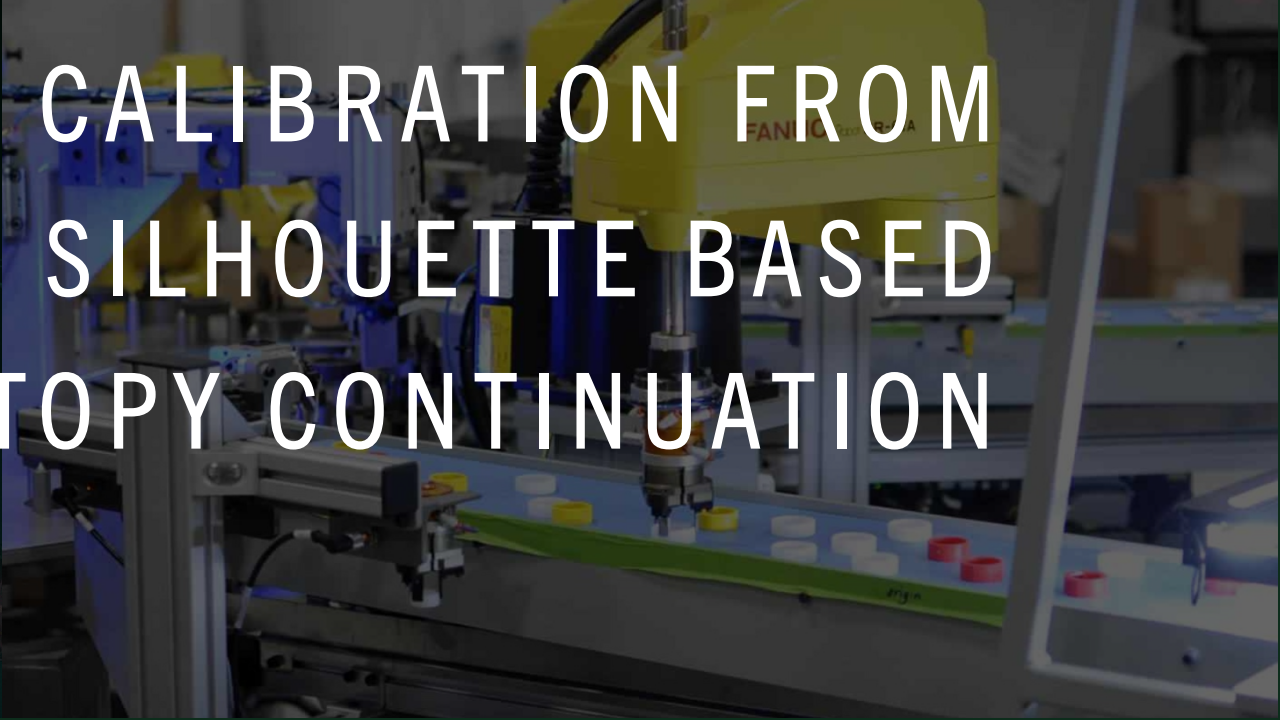
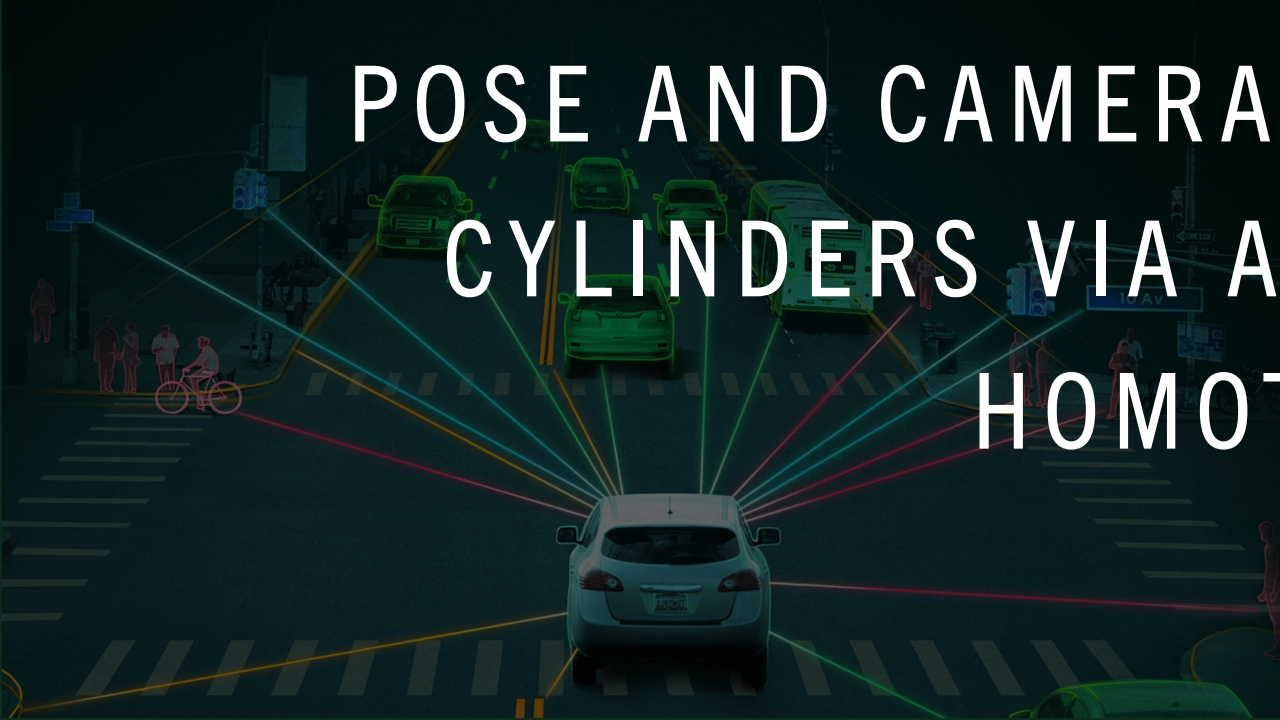
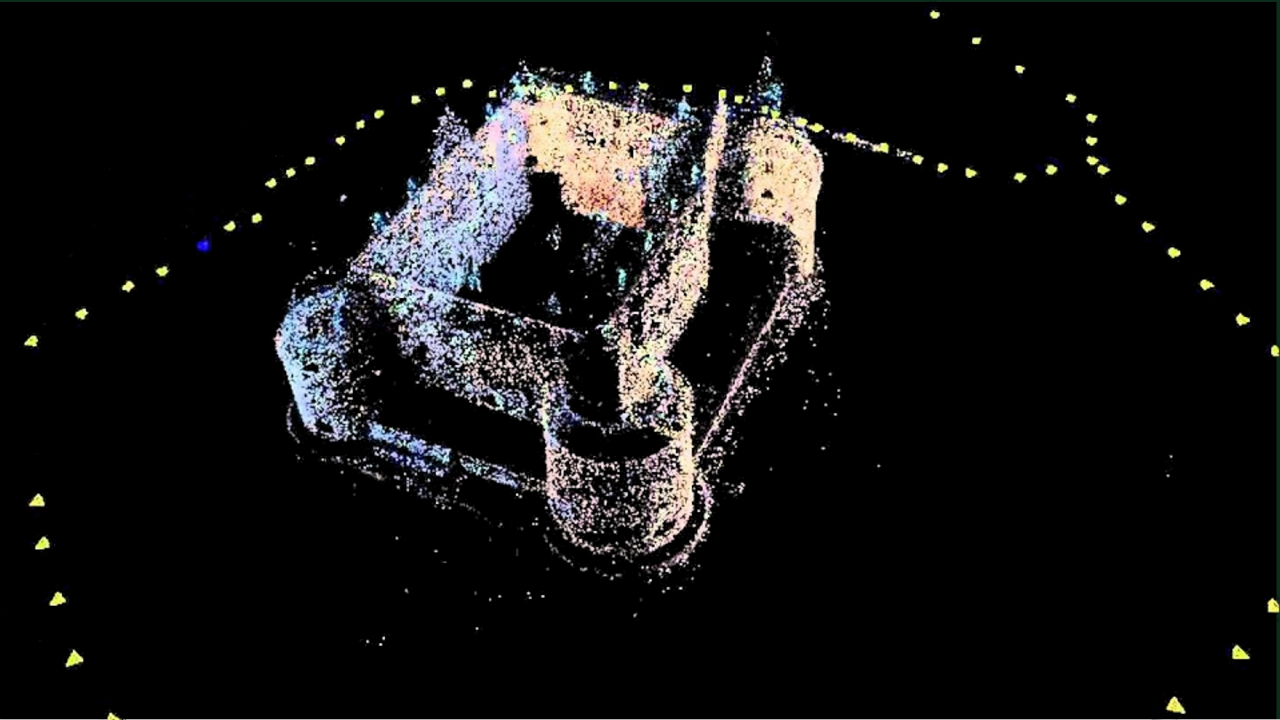
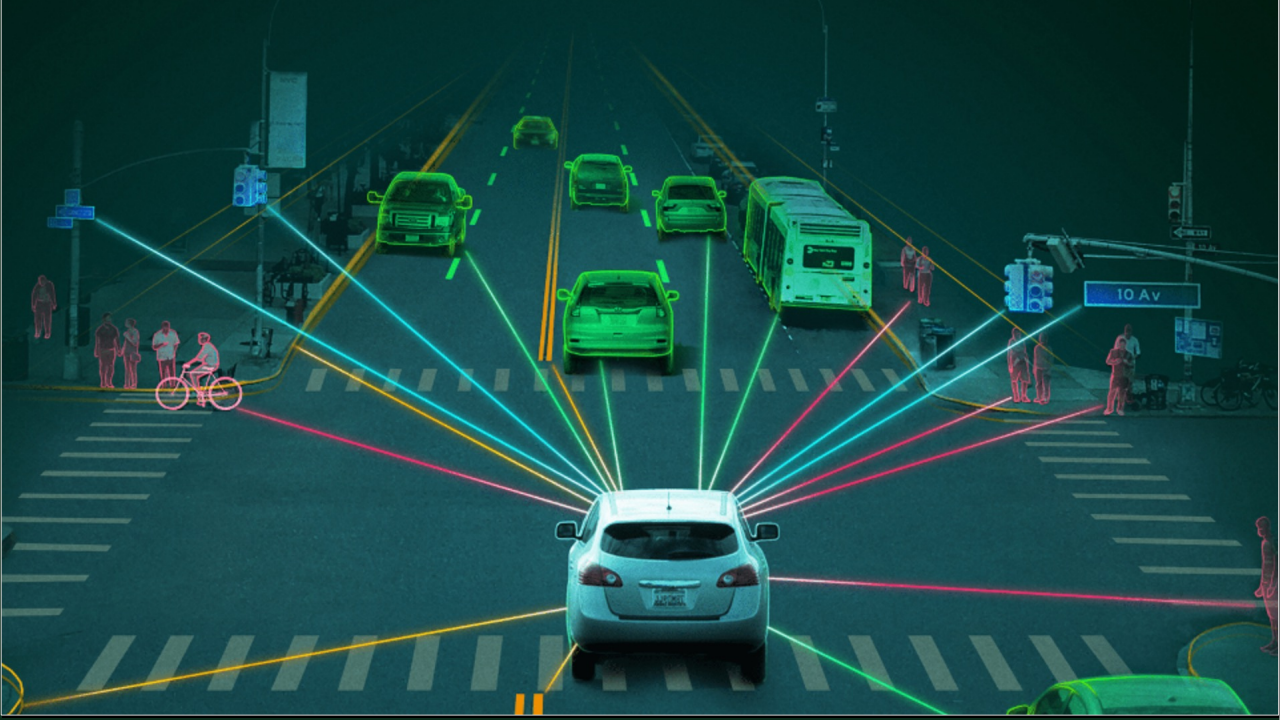




Enrico Barbieri — Luca Magri

POSE AND CAMERA CALIBRATION FROM CYLINDERS VIA A SILHOUETTE BASED HOMOTOPY CONTINUATION





WHERE WE COME FROM: POINTS!

CAMERA MOVEMENT: STRUCTURE FROM MOTION



WHERE WE COME FROM: POINTS!

FEATURES: OCCLUSION



WHERE WE COME FROM: POINTS!

MATCHING

Occlusions: hardly will a point be seen by each camera. Continuity is lost

Scarcity: smooth surfaces are hard to find points on and shiny ones are even view dependent



PROBLEM AT HAND

Trying to achieve

Camera calibration: find the **internal parameters** of the camera

Camera localization: find the **position** and **rotation** of the camera

Possible through

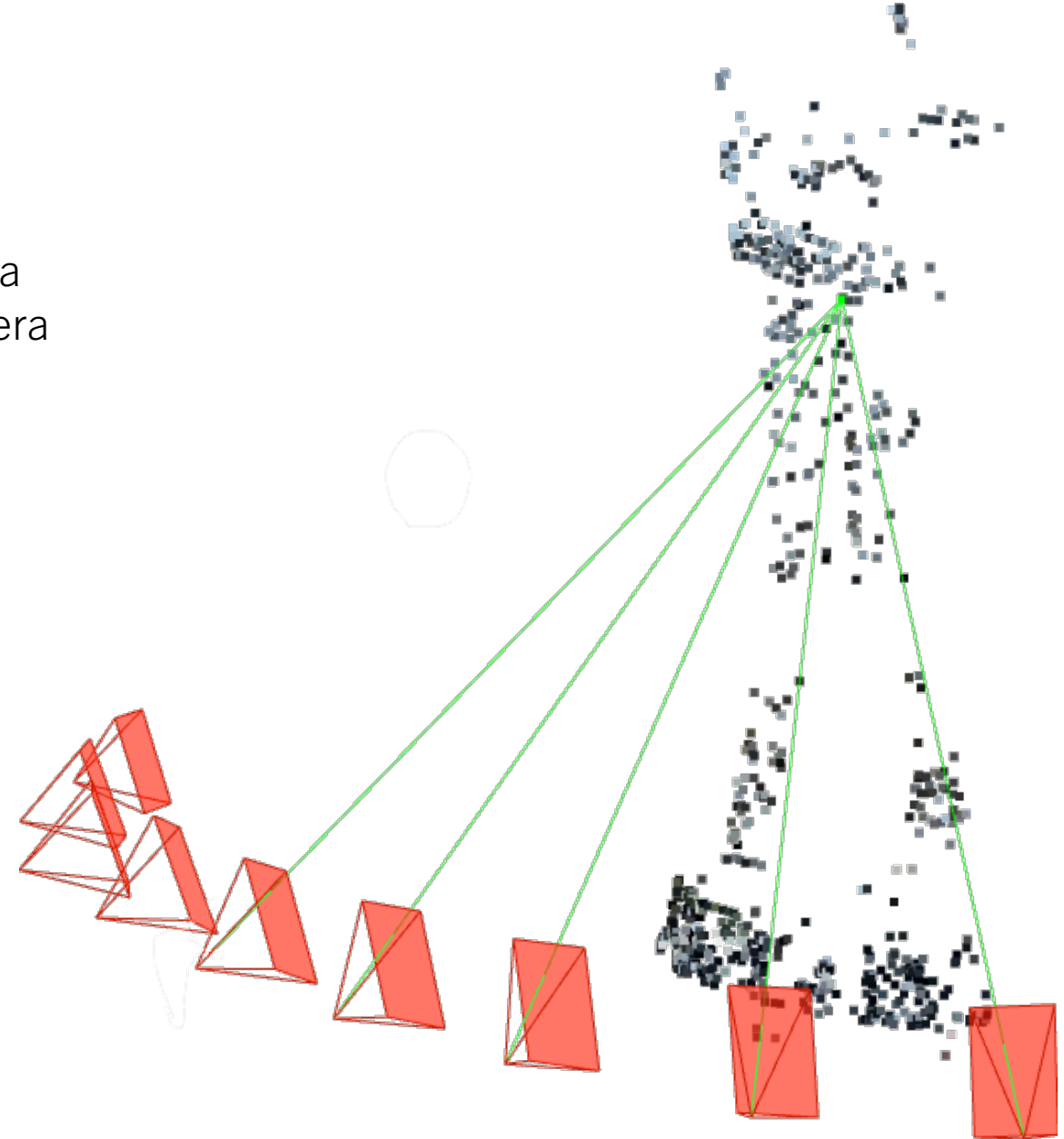
Camera movement: from **multiple images** of the same scene.

Scene landmarks: known point-point, line-plane, **line-cylinder** pairs

Challenges

Textures: scarce high-contrast points

Occlusions: object disappearing during camera movement



WHERE WE ARE HEADED: CYLINDERS

Cylinder contours are visible even in presence of smooth and/or monochrome surfaces

Even with **partial occlusion** (bottom left blue line) a contour could still be identifiable

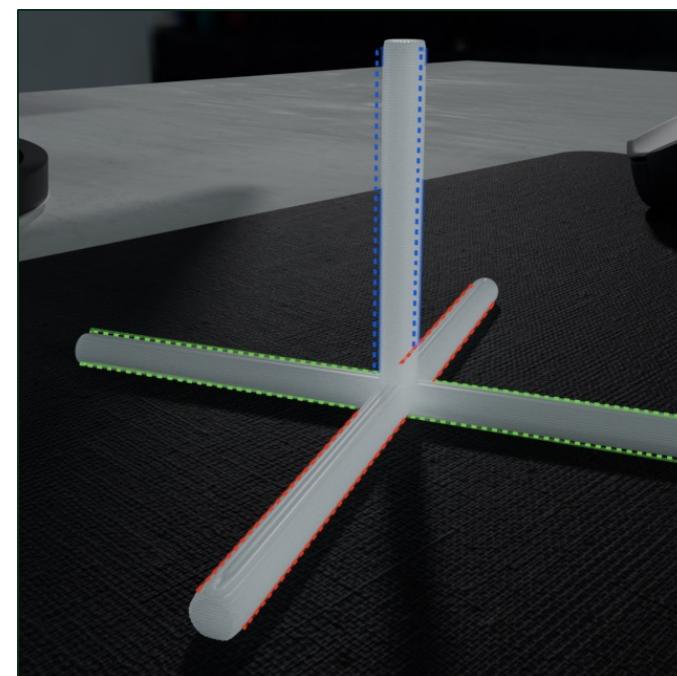
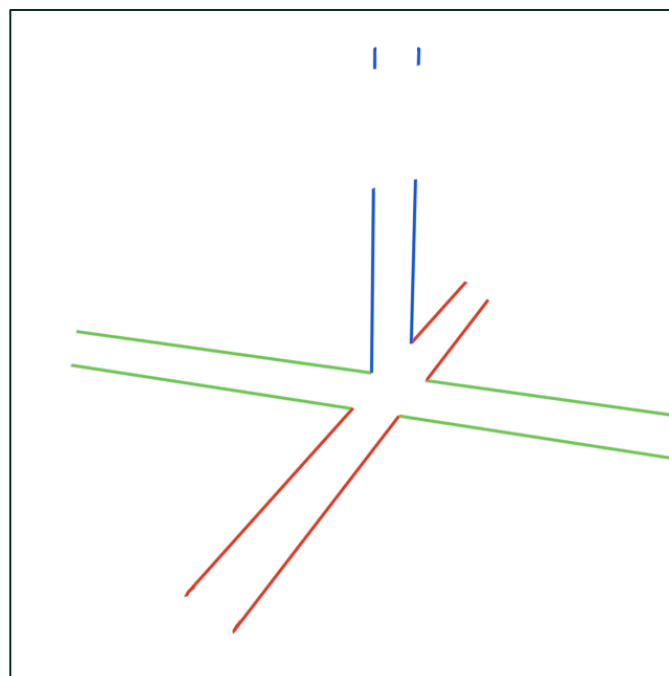
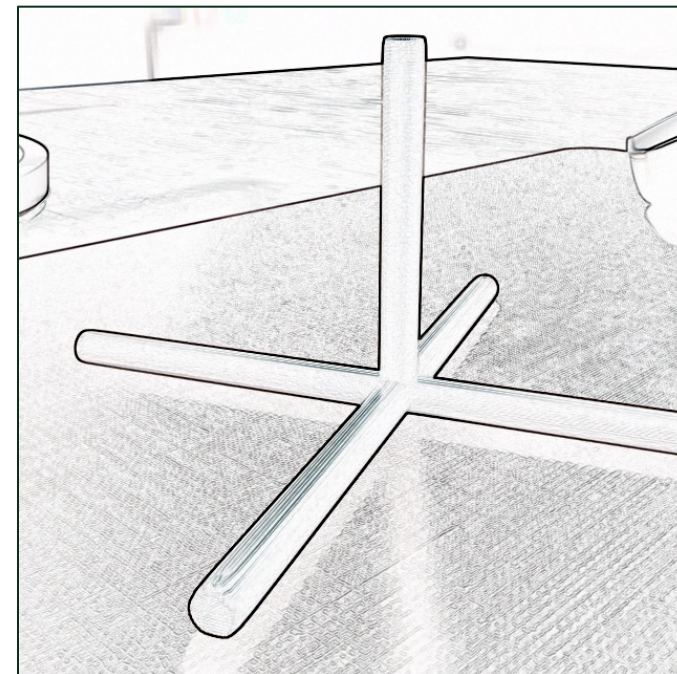
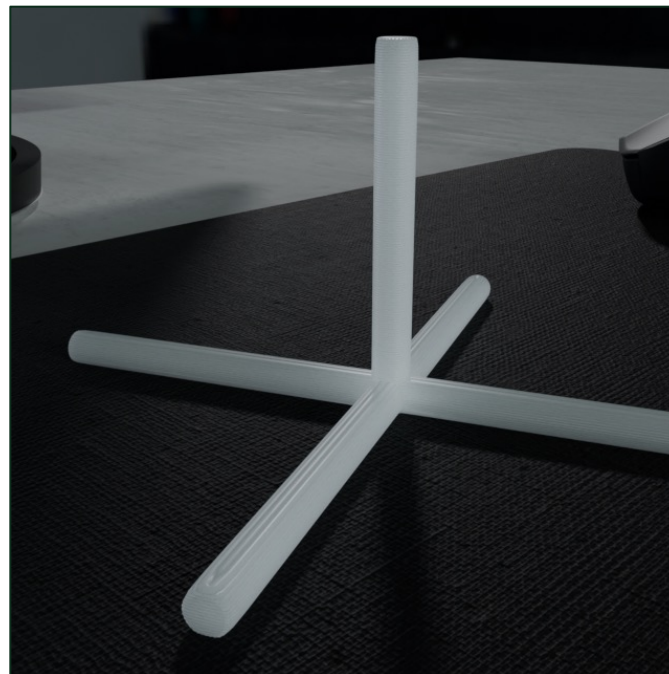
Cylinders **encode more information** and as such less of them are needed

WHAT'S THE CATCH

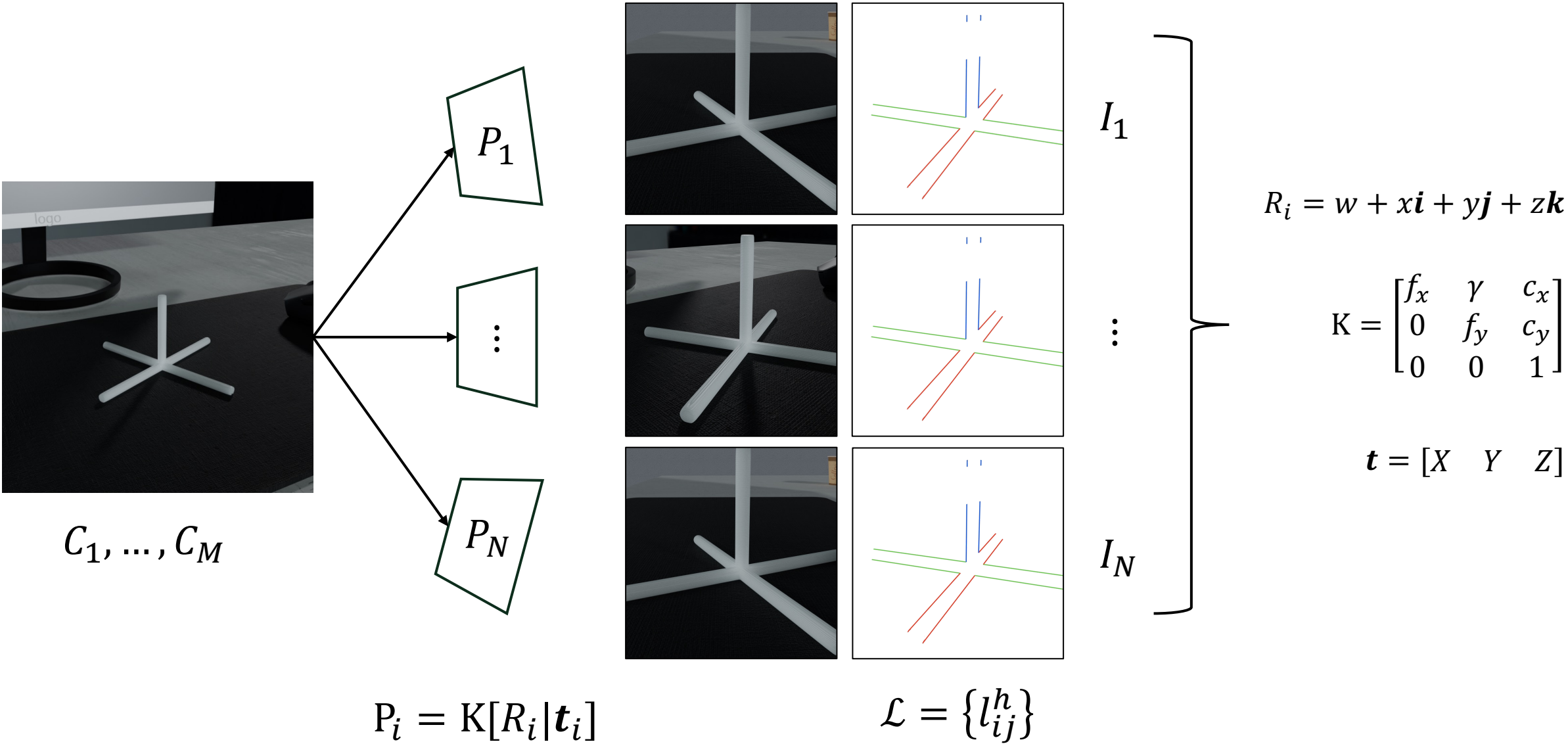
Geometric **problems** arising from cylinders are harder to solve

Distinguishing between **cylinder contours** and other hard surface **lines** is non trivial

Matching pair of lines to the **corresponding cylinder** and be consistent with the matching between shots is also non trivial



PROBLEM FORMULATION

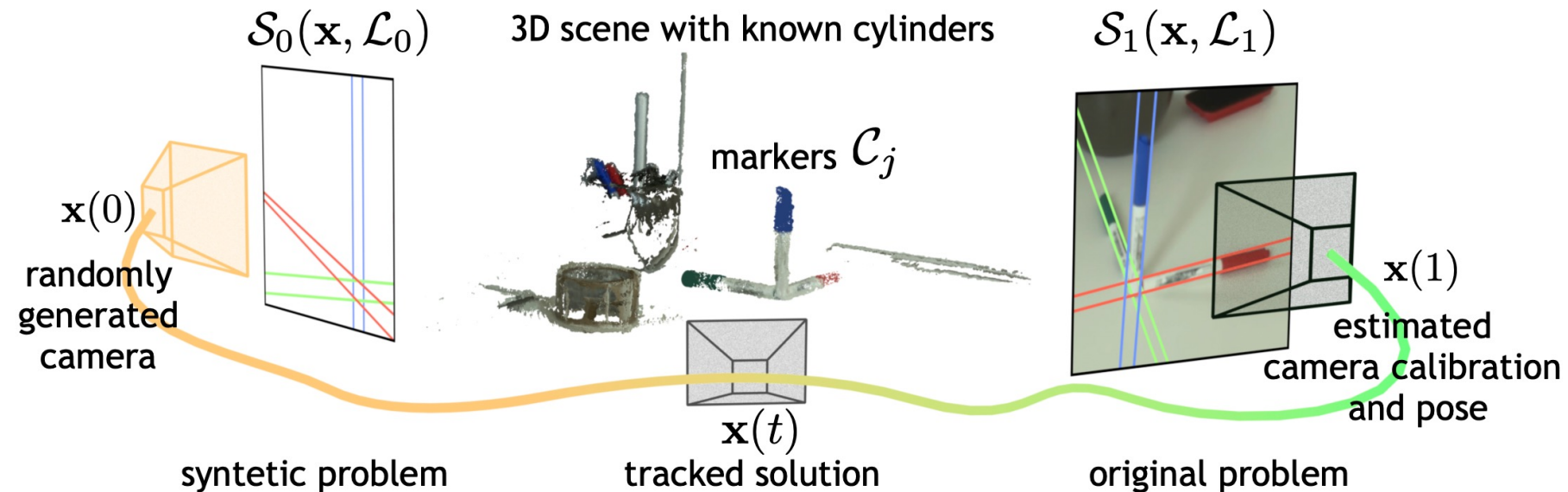


CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION

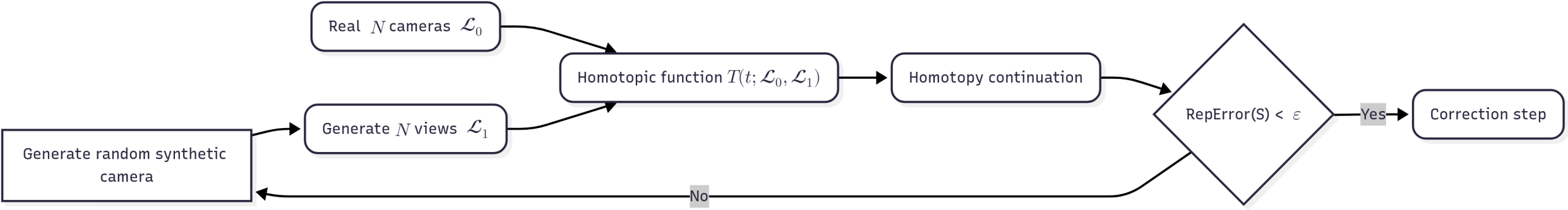
Equation arising from quadric geometry are quartic equations, each with multiple solutions. The **problem** can be **projected** into a higher space in which is linear but is both **computationally intensive**, and **geometrically inconsistent**.

My solution aims to solve both issues by deforming an easy, known version of the problem into the hard, unknown, target problem.

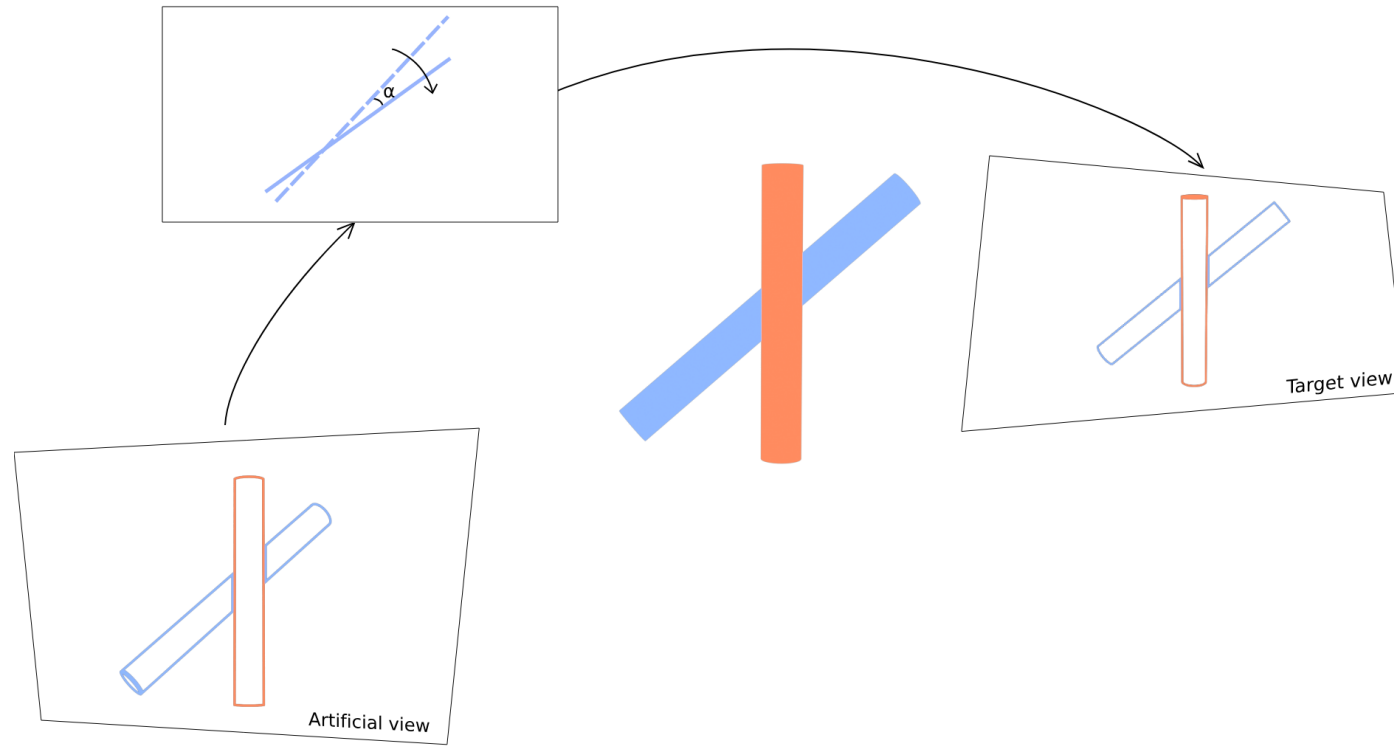
$$\begin{cases} l^h{}^T P Q^* P^T l^h = 0 \\ l^h{}^T P V = 0 \end{cases}$$



CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



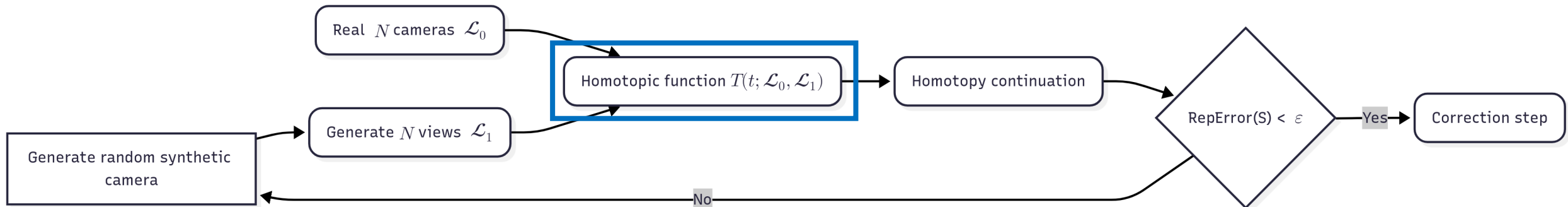
CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



$$\alpha = \widehat{l_0 l_1}$$

$$O = l_0 \times l_1$$

$$T(t; l_0, l_1) = tT_{-O}R_{\alpha}T_O$$



CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION

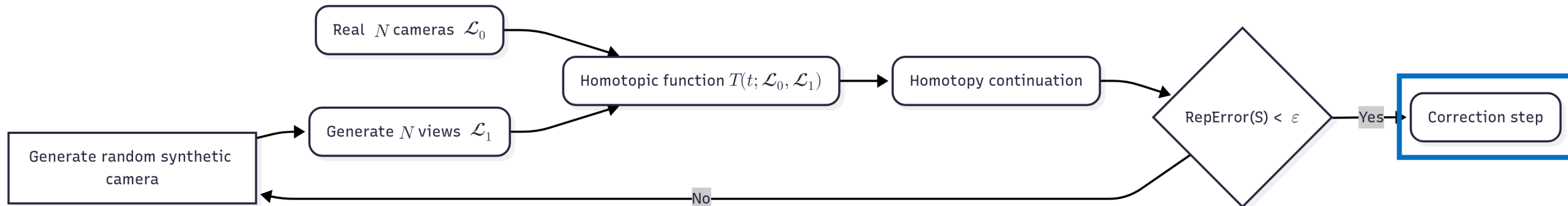
$$l^T K R \mathbf{w} = 0$$

$$l^T K C C^{-1} R \mathbf{w} = 0$$

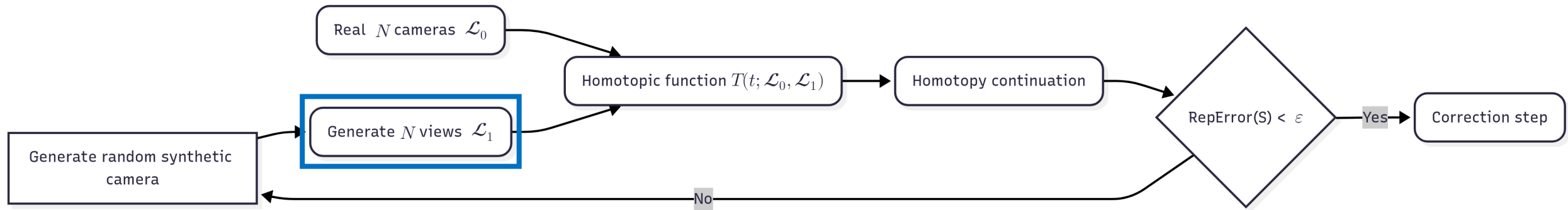
$$C_{f_x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{f_y} = \begin{bmatrix} 1 & 2\alpha/f_x & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{sf_y} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



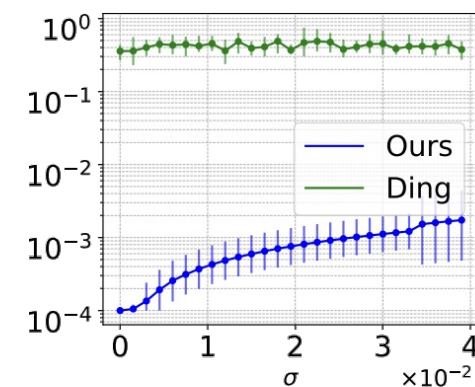
EXPERIMENTS: SYNTHETIC

The proposed method has been compared on **camera localization** against *Gummerson et al.* [1], and on **camera calibration** against *Ding et al.* [2].

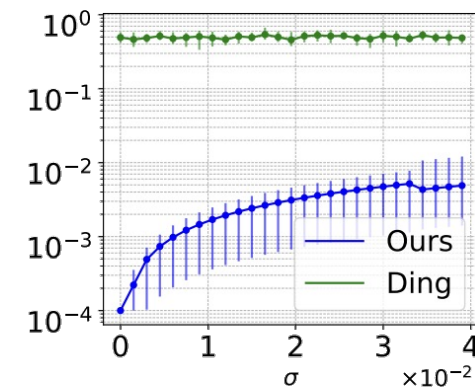
The method **performs orders of magnitude better** than *Ding et al.* [2] on the calibration task.

The method **slightly outperforms** *Gummerson et al.* [1] on **camera localization** while also performing **calibration simultaneously**

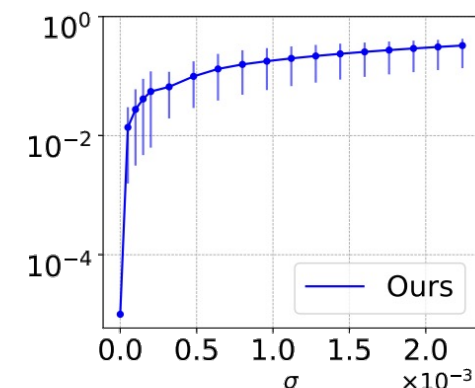
- (1) A. Gummerson, et al. Fast and efficient minimal solvers for quadric based camera pose estimation. In ICPR, 2022.
- (2) D.Ding, et al. Camera calibration based on free-placed single right cylindrical target. IEEE Sensors Journal, 2024.



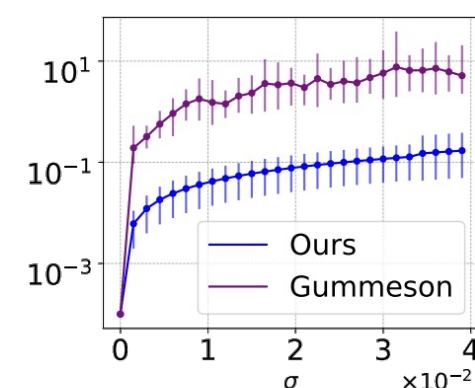
(a) Δf vs. noise.



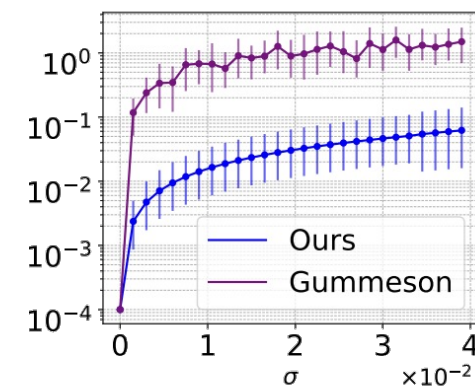
(b) Δc vs. noise.



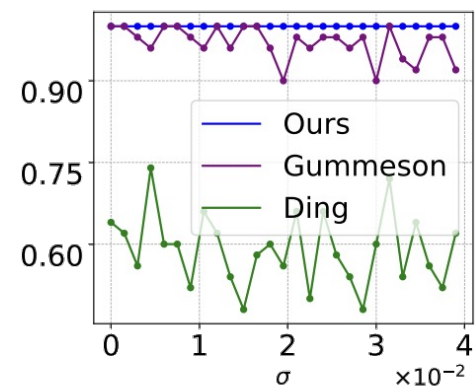
(c) Δs vs. noise.



(d) ΔR (degrees) vs. noise.



(e) Δt vs. noise level.



(f) % of successful trials.

EXPERIMENTS: REAL

Test scene taken from *Gummeson et al.* [1].
Ground truth intrinsics, camera position, and
line positions provided by them.

Our method performs comparably to them
while also **estimating camera** intrinsics
parameters

(1) A. Gummeson, et al. Fast and efficient minimal solvers for quadric based camera pose estimation. In ICPR, 2022.

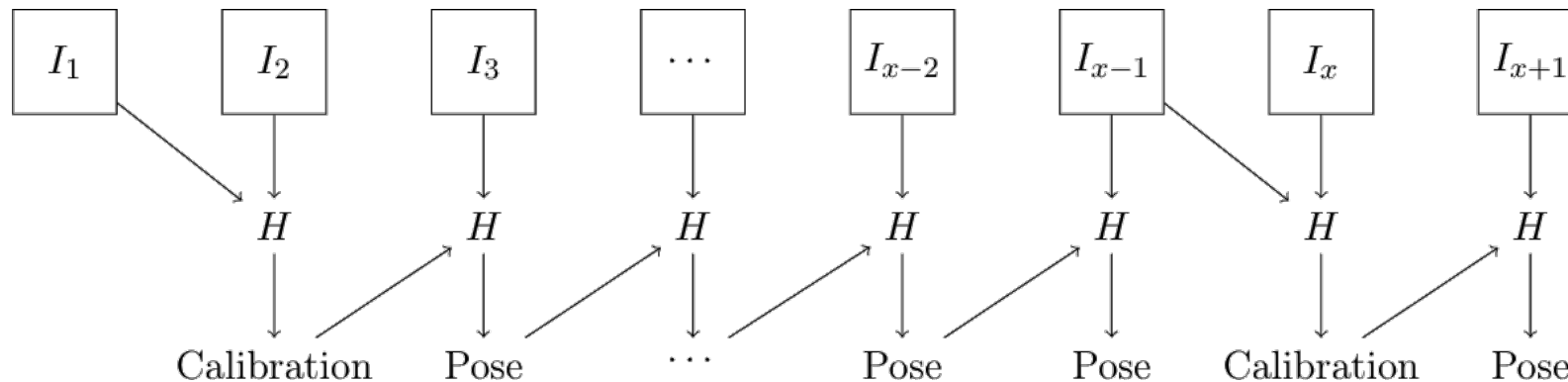


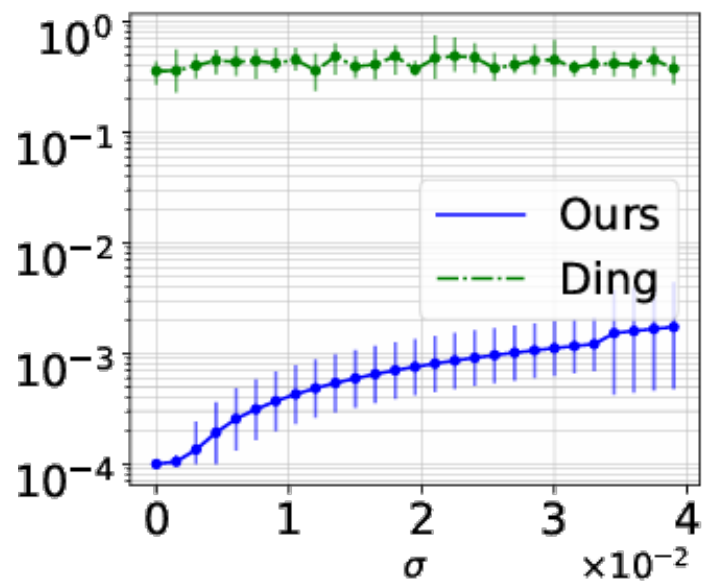
PRACTICAL APPLICATION: SPECIAL EFFECTS

The flexibility of our method allows to seamlessly transition between a calibration and a localization task.

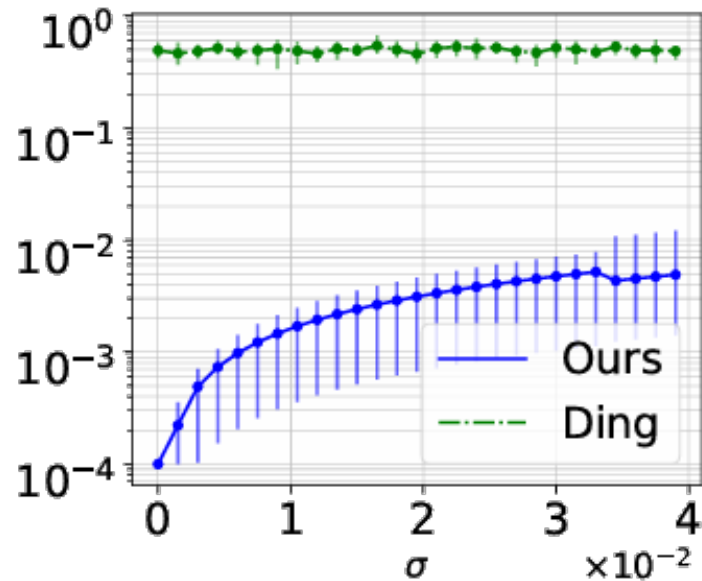
This allows for a localization/calibration feedback loop in which calibration is performed only every N-th frame or if the localization procedure fails.

This allows for fast tracking while the camera is moving while also preserving local consistency

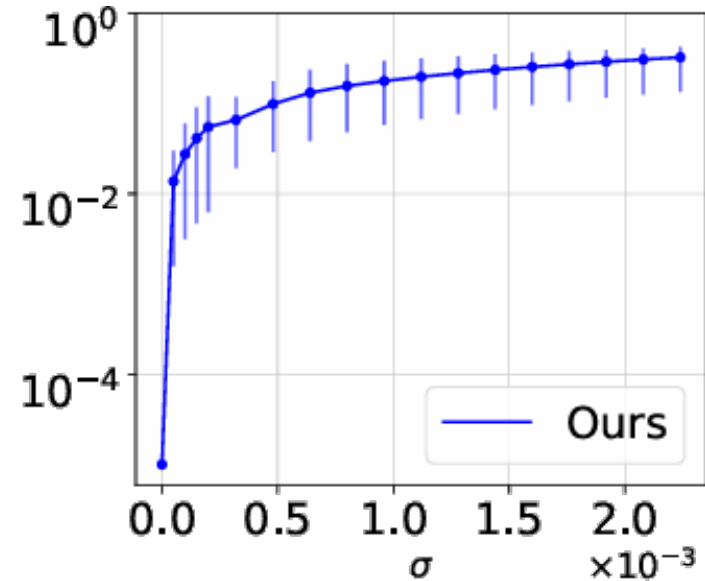




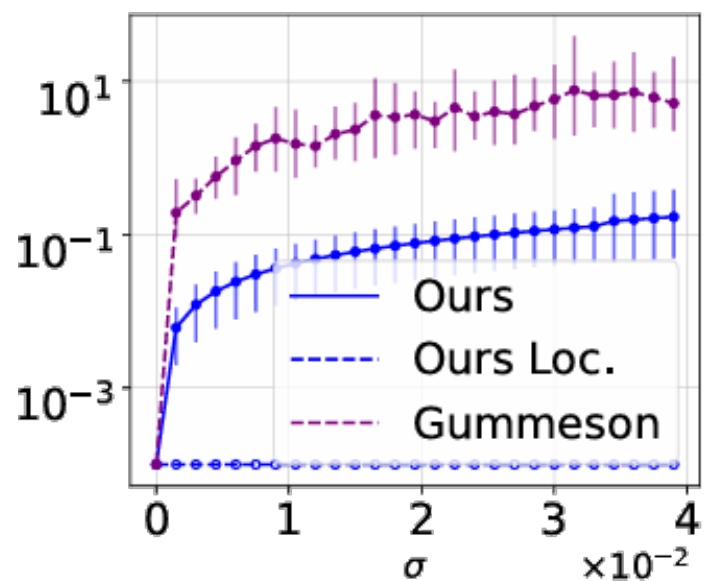
(a) Δf vs. noise.



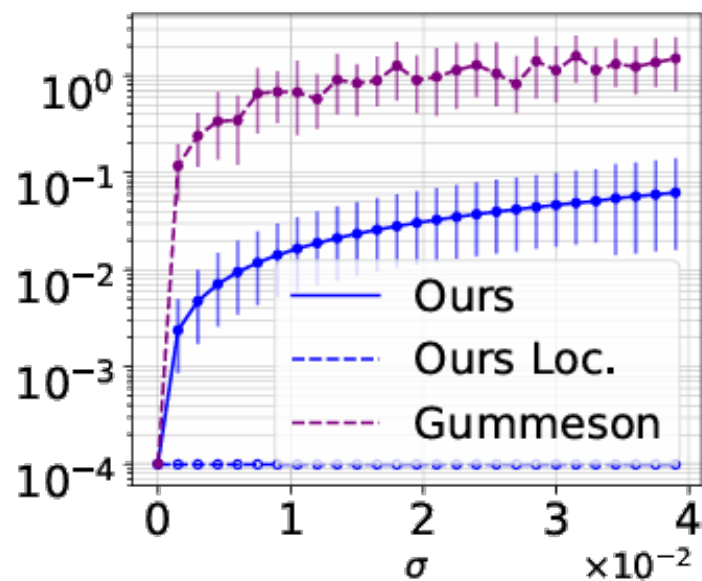
(b) Δc vs. noise.



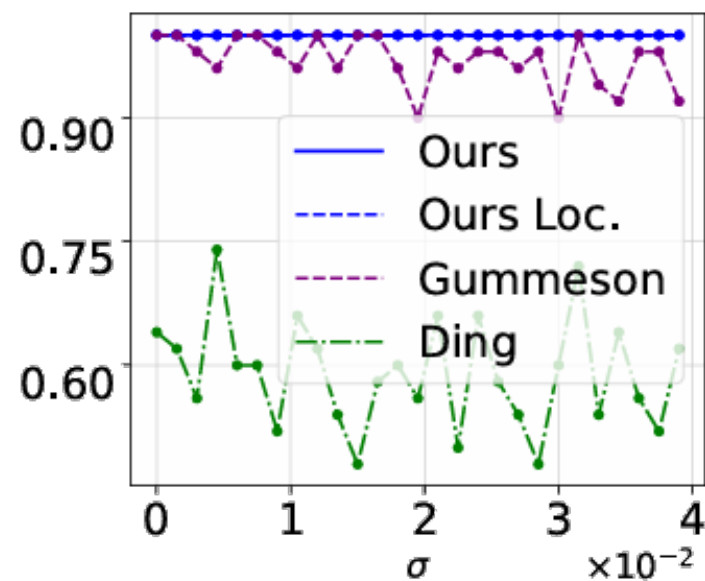
(c) Δs vs. noise.



(d) ΔR (degrees) vs. noise.



(e) Δt vs. noise level.

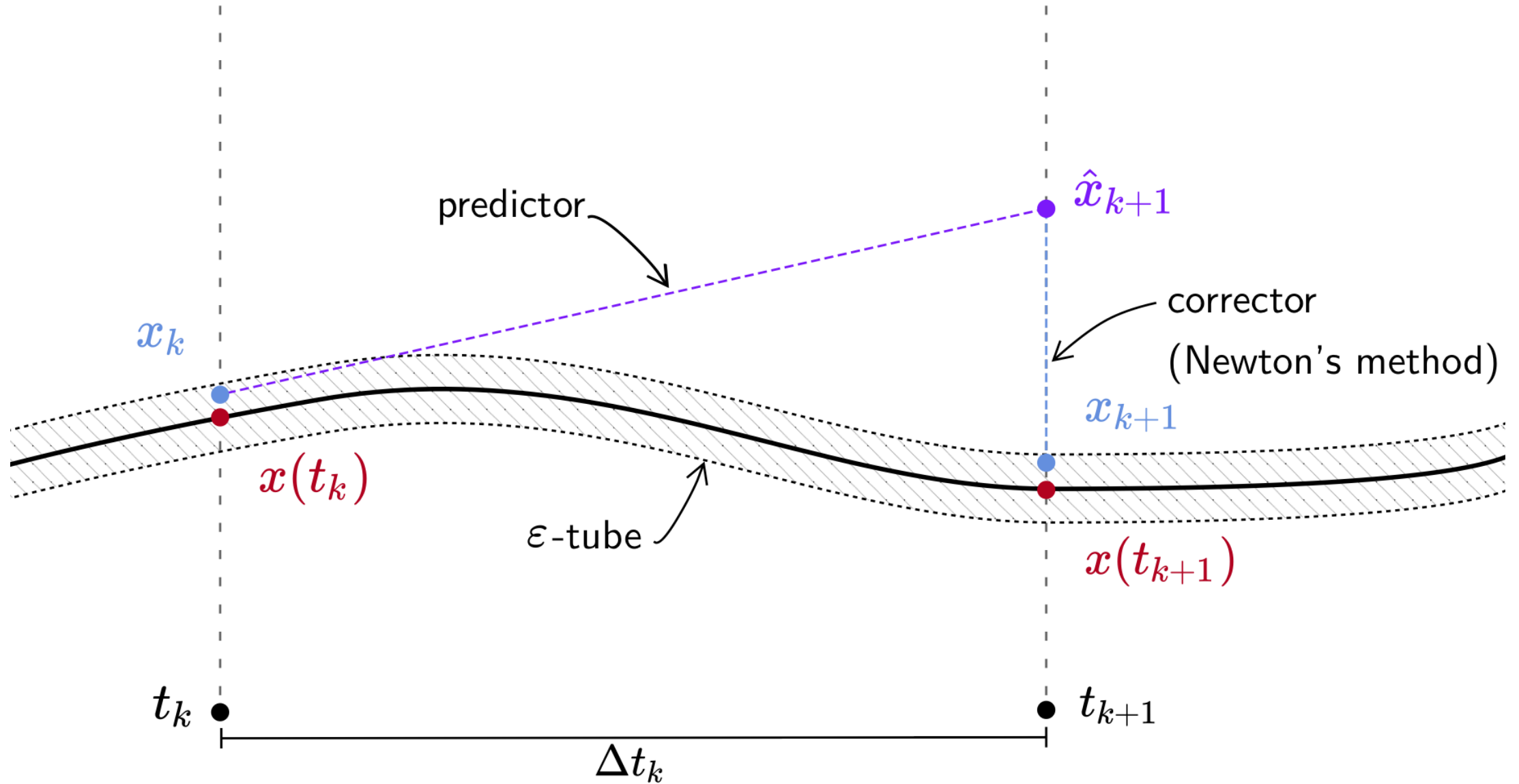


(f) % of successful trials.





HOMOTOPY CONTINUATION – PREDICTOR-CORRECTOR APPROACH



SOLUTION GENERATION